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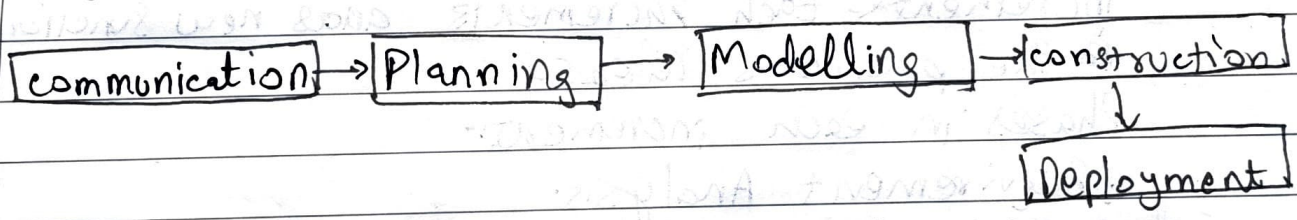
SEPM - Assignment 1

Q.) Explain different software process models?

Soln) Software Process models are structured frameworks that guide the development of software from conception to final deployment. Each model prescribes a different workflow, set of activities, and deliverable structure. The different software process models are:-

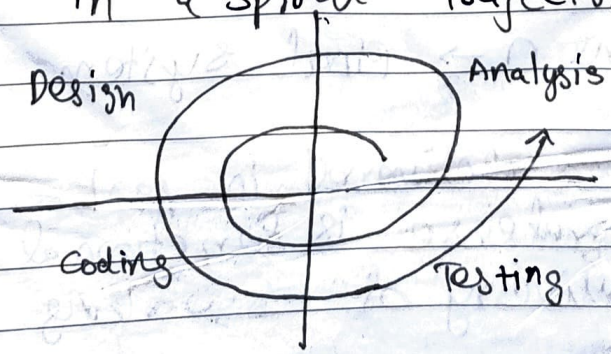
1.) Waterfall model:- The waterfall model is the oldest and most traditional approach. It is a linear-sequential model where the development flows steadily downwards through clearly defined phases. Each phase in a waterfall model must be completed and approved before the next one begins.



Advantages:- It is simple to understand, it has clear milestones to achieve.

Disadvantages:- It is inflexible to changes, working software is produced late.

2.) Spiral Model: The spiral model combines the iterative nature of agile model with the controlled, systematic aspects of waterfall. Its primary focus is risk analysis. The project passes through different phases repeatedly in a spiral trajectory.



The different Phases are:-

→ Analysis: To determine objectives, alternatives and constraints.

→ Design:- Create a blueprint for the architecture of the software.

→ Coding:- Develop ~~the~~ ~~cost~~ and code the application often prototyping.

→ Testing:- Evaluate, deploy and monitor the software.

3.) Incremental Model :- The incremental model is a software process model where the system is developed and delivered in small, manageable increments. Each increments adds new functionality to the previous release.

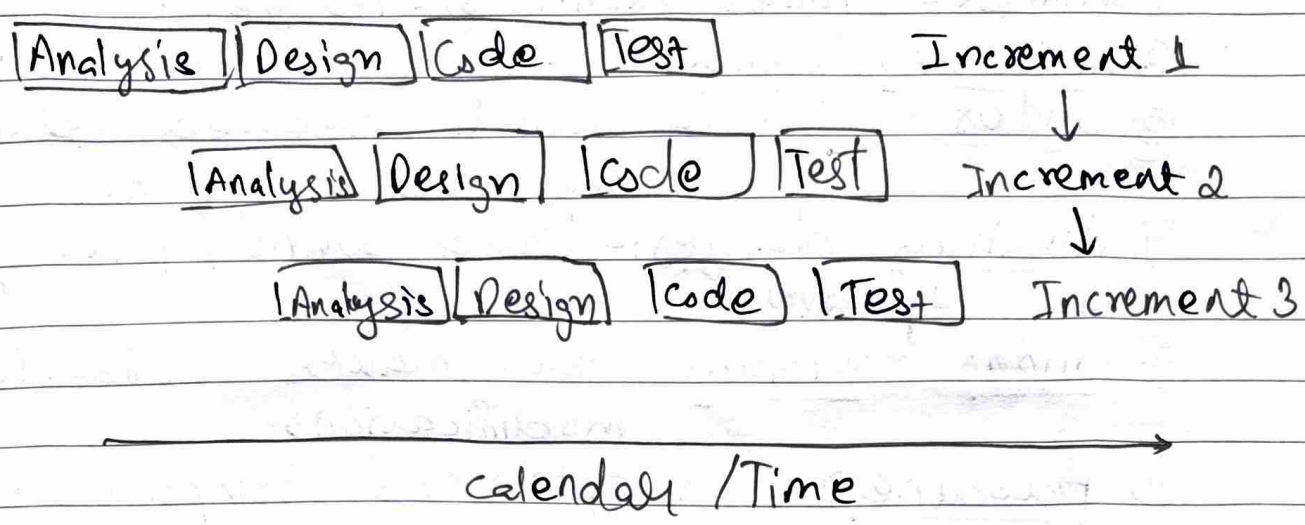
Phases in each increment:-

1.) Requirement Analysis.
2.) Design
3.) Implementation
4.) Testing
5.) Deployment.

Increment 1 → Core functions
Increment 2 → Additional features.
Increment 3 → Advanced features
⋮
Increment n → Final system

In this:

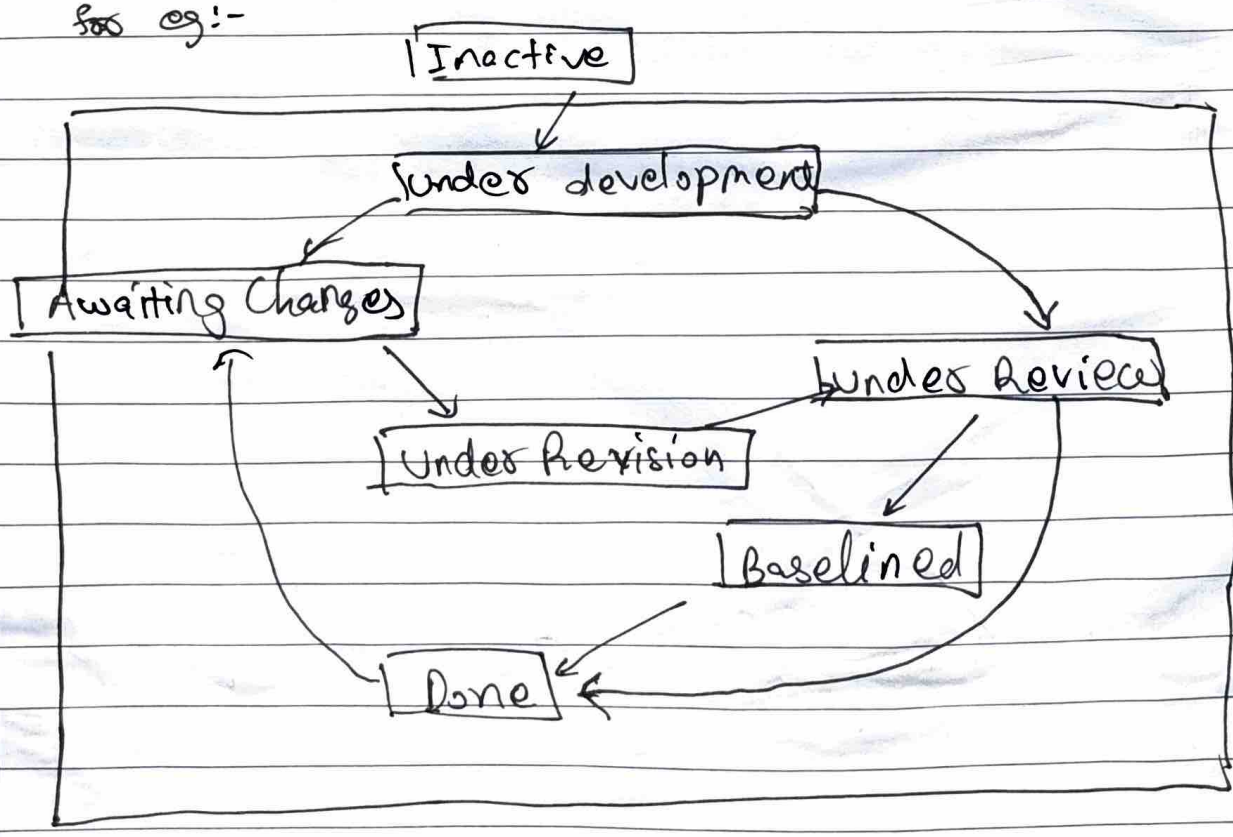
→ Software is delivered in parts
→ Each release is functional
→ Early delivery of working software.



4.) Concurrent Process Model:-

This model allows all SDLC activities to run partially. Each activity is one of several states as soon as something changes in one activity the states of other activities also change. This makes the model flexible and ideal for project where requirements evolve continuously.

for eg:-



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1. Inactive : phase hasn't started yet
2. under development:- work is being created.
3. under review:- work is checked and feedback is given.
4. Awaiting changes:- issues found, phase pauses for updates.
5. under revision:- team makes corrections or modifications.
6. Baselined:- accepted and temporarily proven.
7. done:- May reopen if needed.